

ASearcher-8B: key training setups & profiling status

Agentic-RL-CA rung-3 — branch `sync-partial-rollout-poc`

2026-07-30 04:53 CST (snapshot)

1. Key setups (current values; where set)

Setting	Value	Notes / rationale
Model / data	Qwen3-8B-Base; Base-35k unfiltered; wiki-18/e5 top-5	RL-from-base, no cold start (ASearcher recipe); seed 0
Horizon / context	32 turns nominal; <code>MAX_PROMPT=16384</code> ; <code>MAX_RESPONSE=1024/turn</code>	16k binds first: measured mean ~6 turns, max 26; 24–37% of turns hit the 1024 cap
History	full: past <code><think>+<search>+ 5k-char <information></code> blocks, never dropped	<code>env.asearcher_history</code> ; 5k cap hits 3–8% of retrievals (<code>info_char_cap=5000</code>)
Truncation	<code>clean</code> <code>ctx-terminate</code> (<code>env.ctx_terminate</code> , margin 256); <code>TRUNCATION=left</code> as never-fire valve	episode ends before the model sees a truncated prompt; residual clip 0.5% (prefilter tail — tighten before arms)
Partial rollout	PoC, per-run flag; <code>cycle_turns=8</code> , <code>max_age=4</code> , group-atomic release, per-row <code>policy_version</code> ; no off-policy correction	T1 resume identity byte-exact; invariants exact; 0 stale drops; 1.5–1.6× measured (2.2–2.5× gen)
Prefix caching (vLLM)	ON but defeated by template (step-count precedes history)	only ~200 of ~4.6k mean prompt tokens reusable; template fix = new MDP, deferred to a protocol boundary
Chunked prefill	ON as of <code>grpo_fast</code> + <code>max_num_batched_tokens=17408</code>	fork requires $\geq \text{max_model_len}$ (user-requested 16384 rejected by validation — deviation)
Dynamic batching	<code>DYNBSZ=1: use_dynamic_bsz @ 24,576 tok/GPU</code> (actor/logprob/ref; critic keys via cond) — under test now	replaces fixed <code>MICRO_BSZ=1</code> (~5.2k real tok/forward); gradient-identical; OOM ladder 24576→16384→revert
Batching (fixed)	<code>256×5=1280</code> traj/step; mini 512 rows; <code>remove_padding</code> , <code>balance_batch</code> , grad-ckpt all ON	matched to 4B arms (budget comparability)
Memory posture	KV frac 0.5; critic param+optim offload (gae arms)	actor-only peaked 85GB at KV 0.6; critic OOM'd at <code>wake_up(kv_cache)</code> without offload
Algorithm (fixed)	GRPO vs turnPPO; $\gamma=\lambda=1$; token-mean loss; clip 0.2 (dual 3.0); GRPO KL 1e-3 / turnPPO no KL; entropy 0	§4 of the config audit; unchanged by any speed work

2. Profiling matrix status (04:53 CST)

Run	Status	Key result / ETA
p8bprof_grpo_vanilla	done	7,652 s/step; 5.98 s/traj; 13.3 d/arm
p8bprof_grpo_partial	done	4 cycles; 4.00 s/traj; 8.9 d/arm
p8bprof_ppo_partial	done	3 cycles; 5.76 s/traj; 12.8 d/arm
p8bprof_ppo_vanilla (retry)	running	step 1 in progress under KV 0.5+offload; ETA cfg done ~09–11 CST
p8bprof_grpo_fast	running	DYNBSZ 24576 + chunked prefill + partial; attempt live since 04:27, step 1 ~05:10, preli

Also in flight: 4B p3v2 — token-GRPO **done** (final val@150 **em 0.773**, 3.4 turns); turn-PPO **running** step 73/150 (0.447@50; val@75 next; ETA finish mid-day Jul 30). 8B arm launch: **held** pending **grpo_fast** numbers + your choice (partial/steps/vanilla). Quota 6/6: p3v2-turn-ppo, rung-4×2 (parallel session), ppo-retry2, grpo-fast3.

3. Detailed profiling results (per cycle/step)

config	st	gen(s)	olp(s)	ref(s)	upd(s)	total(s)	rel.traj	mem GB	succ	ent
grpo-vanilla	1	4,722	510	504	2,488	8,247	1,280	85	0.159	2.57
grpo-vanilla	2	3,200	340	340	1,684	7,058	1,280	85	0.229	1.97
grpo-partial	1	553	23	28	125	730	325	70	0.237	2.07
grpo-partial	2	1,753	195	195	967	3,118	700	85	0.184	2.38
grpo-partial	3	2,345	455	452	2,274	5,547	1,255	85	0.170	2.64
grpo-partial	4	1,715	337	339	1,684	5,519	1,445	85	0.197	2.14
ppo-vanilla	1	5,740	503	—	4,850	11,681	1,280	96	0.169	2.57
ppo-partial	1	586	21	—	228	863	290	62	0.217	1.89
ppo-partial	2	1,914	161	—	1,615	3,885	705	77	0.196	2.24
ppo-partial	3	2,254	427	—	4,184	7,373	1,110	77	0.153	2.74

Reading guide: **olp**=old_log_prob recompute, **upd**=actor(+critic) update, **rel.traj**=trajectories trained that step (vanilla always 1280; partial = groups fully terminated). Vanilla step 2 is faster than step 1 because the first update already shortens trajectories (max turns 26→~19). Partial cycles ramp: cycle 1 is fresh-only/cheap, cycles 3–4 are steady state. **ppo-vanilla** has one step (OOM at cycle 2’s KV re-map, pre-fix); **ppo-partial** cycle 4 was cut by the 6 h config timeout. turnPPO **upd** includes the critic; its **ref** column is empty (no KL for that arm — config audit §4). Success/entropy across configs sit in the same band (temp-1.0 sampling noise ±0.04) — no sign that partial rollout moves training metrics at this horizon.

config	trajs	turn-rows	mean turns	succ	5k-cap	1024-cap	cut-pend.
grpo-vanilla	3,584	22,017	6.14	0.195	3.1%	24.3%	0
ppo-vanilla	1,280	7,652	5.98	0.169	7.7%	37.1%	0
grpo-partial	4,712	25,189	5.35	0.187	4.4%	27.0%	107
ppo-partial	4,627	27,494	5.94	0.158	5.1%	27.8%	248

From the per-turn dumps: mean turns 5.3–6.1 everywhere; **5k-cap**= share of retrievals hitting the information-block cap; **1024-cap**= share of turns hitting the response cap; **cut-pend.**= trajectories still pending when the config ended (partial only — excluded from training, never trained truncated).

4. What **grpo_fast** changes (and what it must not change)

Baseline = **p8bprof_grpo_partial** (same data, seed 0, 256×5, 4 cycles, partial args **cycle_turns=8/max_age=4**). Deltas — all **throughput-only**, gradient math untouched:

1. **Dynamic token-budget micro-batching** (DYNBSZ=1): **use_dynamic_bsz=True** at 24,576 tok/GPU for actor update, old_log_prob and ref forwards (critic keys wired for gae arms). Replaces fixed **MICRO_BSZ=1** (~5.2k real tokens/forward). Same minibatch partition (**ppo_mini_batch_size=512**); packing only regroups forwards.
2. **Chunked prefill** + **max_num_batched_tokens=17408** (fork requires $\geq$ **max_model_len**; user-requested 16384 rejected by validation).
3. Nothing else: template, KV 0.5, remove-padding, balance-batch, grad-ckpt, sampling, all algorithm settings identical. OOM ladder: 24576 → 16384 → revert (separately named).

Early step-1 evidence (04:59 CST): update 124.5s→**51.4s**, old_log_prob ~500s-class→**14.6s**, ref→**17.1s** at matched released count (325); gen flat (587 vs 553s — fresh short prompts; the chunked-prefill test is cycle 2’s deep resumed prompts); peak 77.9 GB, no OOM. Known confound vs baseline: baseline ran at KV 0.6, fast at 0.5.

5. Cost history of the fast-config attempts (transparency)

Attempt 1: **set -u** unbound **ADV_ESTIMATOR** in protocol DYNBSZ block (fixed; critic keys moved to **run_condition**). Attempt 2: fork validation **max_num_batched_tokens** $\geq$ **max_model_len** (fixed: 17408 derived). Attempt 3: **running** clean past both failure points. Total overhead ~50 min.

6. Pointers

Config audit: `docs/design/2026-07-30_long_horizon_config_audit.md`. Partial-rollout design: `docs/design/2026-07-29_sync_partia`
Full profiling report: `docs/reports/2026-07-29_p8b_profiling/p8b_profiling_report.pdf`. All docs also at `~/xiaoxuan/arlca-8b/doc`
wandb project [rainyfields/ca-rung3-8b-asearcher](#).